

Cheung Ming Ho Rooney

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EDUCATION

- **University of Toronto - Bachelor of Applied Science and Engineering (B.A.Sc) in Engineering Science (Major in Aerospace Engineering)**
 - **Sep. 2024 - Apr. 2029 (Expected)**
 - **Intended Certificates:** Engineering Business Certificate
 - **Relevant Courses:** Structures and Materials, Vector Calculus and Fluid Dynamics, Praxis I/II/III (Engineering Design and Team Collaboration), Engineering Mathematics and Computation (MATLAB), Introduction to Computer Programming (Python), Computer Algorithms and Data Structures (C), Classical Mechanics, Waves and Modern Physics (Laboratory Skills), Molecules and Materials (Crystal Structures), Probability and Statistics (R), Electromagnetism, Linear Algebra, Ordinary/Partial Differential Equations, Thermodynamics and Heat Transfer, Dynamics, Aerodynamics, Mathematics for Robotics, Economical Analysis, Aerospace Laboratory, Space and Aircraft Flight, Gas Dynamics, Control Systems, Solid Mechanics, Scientific Computing.
 - **CGPA:** 3.98/4.00
- **Richmond Green Secondary School - Ontario Secondary School Diploma**
 - **Sep. 2022 - Jul. 2024**
 - **Highest Academic Standing** for Physics and Accounting.
 - Graduated with average mark **97.7**.

EXPERIENCE

- **Summer Research Student** [*Advanced Modelling and Hybrid Simulation Research Group*] May-Aug 2026
 - **Received** funding from the Undergraduate Student Research Award.
 - **Developed** a web hosted relay architecture for communication between a publicly hosted server on AWS, substructures and solvers using TCP/IP protocols in **Python**.
 - **Assisted** graduate and PhD students on preparing and conducting civil engineering experiments.
 - **Organized and Presented** progress and development in weekly slideshow reports.
- **Data Analysis and Engineering Intern** [*MTR Corporation Limited*] Jul-Aug 2025
 - **Inspected** track irregularities and conducted on-site investigations during night shifts.
 - **Developed** software-based algorithms using **Python** and **Excel VBA** alongside simple mathematical models to detect track irregularities from measurement data.
 - **Generated Graphs and Visuals** to pinpoint critical areas and presented the information to my supervisor using **infographics**.
- **Private and Volunteer Tutor** 2022 - Pres.
 - **Tutor** for physics, biology, chemistry, advanced functions and calculus for the Ontario curriculum.
 - **Summarized** course information and created practice exercises for students to engage in tutorial sessions.

RELEVANT PROJECTS

- **Holdover Fire Monitor System** [*Praxis III, University of Toronto, Toronto, ON*] Jan-Apr 2026
 - **Prototyped** a physical screw-shaped detector using 3D modelled components, Raspberry Pi, different metric sensors and **Python**.
 - **Designed** an IoT system that can pick up signals from any detector and display them via the internet using **Grafana**.

- **Recorded** team development progress, versions and meetings in a structured and accessible design dossier.
 - **Presented** results using posters, slideshows and presentations with demos.
- **TimeTabler** [Github] Feb 2026
 - **Designed** a **Python** script that intakes To-Do txt Sfiles and output a complete calendar using **tabulate** and **datetime** modules.
 - **Formulated a Mathematical Model** to quantify the days-due factor and priority/difficulty of each task to help the user decide which task to pursue first.
- **PokeStudy** [Github] Jan 2026
 - **Developed** algorithms and applied dictionary data structures in **JavaScript** to extract Pokemon data from APIs.
 - **Learned basics of Web Design** through **HTML and CSS** in order to produce Pokeball patterns and generate animations.
- **Sorteria** [Praxis II, University of Toronto, Toronto, ON] Jan-May 2025
 - **Researched** current practices and proposed implementations for solving right-hook collision scenarios in Downtown Toronto. Specifically regarding the viability of **Computer-vision, Sensor and Infrastructure** designs on bicycles.
 - **Presented ideas** to project head and key stakeholders, including proposing design ideas with **On-Shape**.
 - **Visualized** designed ideas on a presentation poster and one-pager for the general public.

EXTRA-CURRICULAR ACTIVITIES

- **Operations Director for Club for Biomedical Engineering (CUBE)** Sep 2025 - Pres.
 - **Organized** events for the club, including room booking, resources and etc.
 - **Recorded** inventory of the club using **Google Sheets**.
 - **Lead** the involvement of Junior Executives using a responsibility rotation system.
- **Intermural Basketball Team member (Skule D)** Jan 2026 - Pres.
 - **Committed** to weekly team games on Wednesday nights.

TECHNICAL SKILLS

- **Languages:** Python, C (with Arduino), JavaScript, HTML, CSS, RISC-V Assembly, MATLAB, SystemVerilog, Excel VBA.
- **CAD:** Onshape, Fusion 360.
- **Laboratory Design:** Handling lab data, apparatus and identifying error sources.
- **Team management:** Designing timetables for team members, planning work equally and accordingly for team members using Google/Outlook Calendar.
- **Data Collection and analysis:** Using Excel/ Google Sheets with built in formulas to organise and extract critical information.
- **Graphics Design:** Using suitable graphs or clear and easy-to-read designs to convey key project ideas.